Fermentation-derived / single-cell protein ingredients (precision-fermented, cultured, yeast-biomass proteins): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

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PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified human-grade standard for precision-fermented, cultured, and single-cell / yeast-biomass protein ingredients used in pet food, stated as sold on a dry powder basis. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

1 | How every number on this standard is set

This subcategory covers a precision-fermented / single-cell / cultured PROTEIN INGREDIENT (spray-dried, heat-inactivated microbial biomass, e.g. Saccharomyces cerevisiae expressing an animal-identical protein) sold as a constituent of pet food. It is legally FEED (FDA CVM review, AAFCO, EU Directive 2002/32). Two read-across facts set the number honestly. First, a single-cell-protein biomass is not contamination-free by construction: anionic cell-wall groups biosorb Pb/Cd/Ni/Cr, arsenate is actively imported through the yeast phosphate transporters, and the metal load tracks the fermentation feedstock (plant sugars/molasses) and the mineral/trace-element premix — so Cd, Pb, and inorganic arsenic are real, load-bearing risks and are never marked not-material. Second, the strictest applicable HUMAN-FOOD analog on a dry as-sold basis is the plant-protein fraction (cereals, pulses, legumes), the same strict analog set the finished pet food already reads across; adopting it (rather than the looser same-class spec) keeps the ingredient limit at least as strict as any finished food it enters (Method v2.0 strictest-applicable rule, and the anti-circumvention guardrail against a looser ingredient tier laundering into a stricter finished food). The resulting toxic ceilings (Pb 20, Cd 40, tAs 100, iAs 100, tHg 20 ug/kg) are stricter on every toxic than the only governed same-class human-food spec, the EU novel-food Yarrowia lipolytica yeast-biomass Union-list contaminant limits (Pb 100, Cd 100, As 150, Hg 100 ug/kg; Reg (EU) 2019/760 amending 2017/2470), which is disclosed as the corroborating over-compliance

floor and never selects the number. The applicable feed floor is disclosed on every row and never selects the number.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	20	human-grade read-across (Hong Kong)
Cadmium (Cd)	Tier 1	40	human-grade read-across (European Union)
Total arsenic (tAs)	Tier 2	100	human-grade read-across (Paraguay)
Inorganic arsenic (iAs)	Tier 1	100	human-grade read-across (Israel)
Mercury, total (tHg)	Tier 1	20	human-grade read-across (Hong Kong)
Methylmercury (MeHg)	Tier 1	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	Tier 2	800	human-grade read-across (European Union)
Tin (Sn)	Tier 2	250,000	human-grade read-across (Singapore)
Aluminium (Al)	Tier 2	100,000	human-grade read-across (China)
Total chromium (Cr)	Tier 2	500	human-grade read-across (Hong Kong)
Hexavalent chromium (Cr-VI)	Tier 2	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION

Lead (Pb)	20	Hong Kong — Cereal-based food intended to be consumed principally by persons under the age of 36 months
Cadmium (Cd)	40	European Union — pulses, general
Total arsenic (tAs)	100	Paraguay — pulses, dried legume seeds, except soybean
Inorganic arsenic (iAs)	100	Israel — Dried legumes, including peanuts and soybeans
Mercury, total (tHg)	20	Hong Kong — Rice, husked rice, polished rice, maize, maize flour, wheat, wheat flour
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	800	European Union — cereals except durum/rice/husked-rice/-pseudocereal/oat rows
Tin (Sn)	250,000	Singapore — human-food analog
Aluminium (Al)	100,000	China — human-food analog
Total chromium (Cr)	500	Hong Kong — Vegetables, other than pulses
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

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References

[1] China — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Cd, Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Hong Kong — human-food maximum levels
Binds Pb, tHg, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] Israel — human-food maximum levels
Binds iAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[5] Paraguay — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[6] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/pet--food-fermentation-derived-protein. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-pet-food-fermentation-derived-protein.